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The Hitchhiker's Guide to Markup Estimation: Assessing Estimates From Financial Data

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1 Big picture

- **Question.** Can macroeconomists learn anything reliable about firm-level markups from financial statements, even though those data do not contain firm-level prices or comparable physical quantities?
- **Benchmark concern.** The production approach to markup estimation needs an output elasticity for a flexible input. When researchers only observe revenue rather than quantity, production-function estimation is generally biased because prices sit in the residual.
- **Core answer.** The paper shows that this bias usually *does not destroy all information*. Revenue-based markups are generally bad for measuring the *average level* of the markup, but they can still be quite informative about *dispersion across firms* and *trends over time*.
- **Why this is subtle.** Revenue-based markups are useless only in the knife-edge case where the estimated elasticity collapses all the way to the *revenue elasticity*. In most practical settings, the estimated elasticity is instead a biased version of the *output elasticity*, so markup variation survives.
- **Main theoretical mechanism.** The bias depends on the covariance between firms' prices and the instruments used in production-function estimation, typically lagged inputs. If prices are orthogonal to those instruments, revenue-based markups can track true markups closely.
- **Three layers of evidence.** The paper combines: (i) an analytical framework, (ii) Monte Carlo simulations of an Atkeson–Burstein style oligopoly model, and (iii) French administrative data with both prices and quantities.
- **Main quantitative simulation result.** In the practical case with revenue data but no price data, estimated markups still correlate 0.94 with true markups in the calibrated model.
- **Main empirical result.** In French manufacturing data, the within-sector correlation between revenue- and quantity-based markups is about 0.61 in levels and 0.80 in first differences.
- **Aggregate implication.** Revenue data mismeasure the *level* of aggregate markups, but the *time-series trend* in aggregate markups is fairly robust.
- **Why the paper matters.** It rescues a large macro/trade literature from the strong claim that accounting-data markups are always uninformative. The correct lesson is narrower: use revenue-based markups for relative comparisons and trends, not for disciplining average markup levels.

2 Data and measurement (key objects)

2.1 Three empirical environments

- **Analytical environment.** The theory begins with a stripped-down firm problem: one variable input, a production function, and imperfect competition that makes revenue differ from physical output.
- **Simulation environment.** The Monte Carlo exercise embeds the theory in a quantitative oligopoly model with heterogeneous firms, translog production, aggregate shocks, and measurement error.

- **Administrative data environment.** The empirical exercise uses French manufacturing firms for which the authors can construct both quantity-based and revenue-based markups and compare them directly.

2.2 French production and financial data

- **Sources.** The paper merges FARE (firm-level financial statements) with EAP (firm-product quantities and revenues).
- **Period.** 2009–2019.
- **Coverage.** FARE covers the universe of non-financial firms via tax filings. EAP is a manufacturing survey covering firms with at least 20 employees or 5 million euros in revenue.
- **Price construction.** For each firm-product, price is revenue divided by quantity. Unit values are standardized by the revenue-weighted average price of the 10-digit product.
- **Firm-level price.** If a firm sells multiple products, its price is the sales-weighted average of those standardized product prices.
- **Firm-level quantity.** Quantity is recovered as firm revenue divided by the firm-level standardized price.
- **Other variables from FARE.** Revenue, wage bill, capital, purchased services, and purchased materials.
- **Sample restrictions.** Observations with missing, zero, or negative core variables are dropped; nominal variables are deflated with EU-KLEMS deflators.
- **Empirical scale.** The final empirical analysis uses roughly 147,000 firm-year observations after trimming and common-sample restrictions.

2.3 Why materials are the variable input

- The production approach needs a flexible input chosen cost-minimally each period without dynamic adjustment costs.
- In the empirical implementation, **materials** play that role. The output elasticity of materials is combined with the material revenue share to back out markups.
- The authors also report how markups correlate with the profit rate, labor share, material share, and market share, because those are the main downstream uses of these estimates in macro work.

2.4 Simulation calibration and objects

- **Model.** Atkeson and Burstein (2008)-style double-nested CES demand with Cournot competition.
- **Production.** Firms share a translog production function but differ in productivity and in the endowment of a fixed input.
- **Shocks.** Both firm-level and aggregate shocks are present; observed output includes measurement error.

- **Monte Carlo design.** 200 simulations, each with 1,600 firms, 180 markets, and 40 periods.
- **Purpose.** This is the clean validation environment because true markups are known by construction.

2.5 Main empirical markup objects

- **Firm-level quantity markup.** Uses the estimated output elasticity from quantity data.
- **Firm-level revenue markup.** Uses the same production-approach formula, but with output elasticities estimated from revenue data.
- **Aggregate markup.** The preferred aggregate statistic is the sales-weighted harmonic mean of firm-level markups, which is the welfare-relevant object in a broad set of models.

3 Models and empirical specifications (all equations + interpretation)

3.1 The production approach to markup estimation

The core Hall–De Loecker–Warzynski identity is

$$\mu_{it} = \alpha_{it} \frac{P_{it} Y_{it}}{W_t V_{it}}, \quad \alpha_{it} \equiv \frac{\partial Y_{it}}{\partial V_{it}} \frac{V_{it}}{Y_{it}}. \quad (1)$$

- **Interpretation.** Markups equal the ratio of a flexible input’s output elasticity to that input’s revenue share.
- **Data requirement.** The revenue share $W_t V_{it} / (P_{it} Y_{it})$ is observable in accounting data.
- **Main difficulty.** Researchers must estimate α_{it} , and that is exactly where revenue-versus-quantity problems arise.

3.2 Why revenue-based markups can still contain information

Suppose firms face inverse-demand elasticity ε_{it} , so that the static pricing rule is $\mu_{it} = (1 - \varepsilon_{it})^{-1}$. Then the revenue-based markup estimator can be written as

$$\hat{\mu}_{it}^R \equiv \hat{\alpha}_{it} \frac{P_{it} Y_{it}}{W_t V_{it}} = \frac{\hat{\alpha}_{it}}{\alpha_{it}(1 - \varepsilon_{it})} = \frac{\hat{\alpha}_{it}}{\alpha_{it}} \mu_{it}. \quad (2)$$

- **Key insight.** Revenue-based markups fail only if $\hat{\alpha}_{it}$ collapses to the revenue elasticity $\alpha_{it}(1 - \varepsilon_{it})$.
- **If $\hat{\alpha}_{it} = \alpha_{it}$.** True markups are recovered exactly.
- **If $\hat{\alpha}_{it} = \alpha_{it}(1 - \varepsilon_{it})$.** Markups become mechanically equal to 1 and contain no information.
- **Typical case.** $\hat{\alpha}_{it}$ lies somewhere in between, so true markup variation survives.

3.3 Baseline Cobb–Douglas production function

The simplest analytical production function is

$$y_{it} = \alpha v_{it} + \omega_{it}. \quad (3)$$

- **Interpretation.** Log output is linear in the variable input, with productivity ω_{it} observed by the firm but not the econometrician.
- **Identification under quantity data.** Lagged input v_{it-1} can identify α if it is relevant and excluded from the current productivity shock.

3.4 The omitted-variable bias from using revenue instead of quantity

If one runs a quantity-style IV-GMM estimator on revenue, the probability limit of the estimated elasticity is

$$\hat{\alpha} = \alpha + \frac{\mathbb{E}[p_{it}v_{it-1}]}{\mathbb{E}[v_{it}v_{it-1}]}. \quad (4)$$

- **Interpretation.** Prices enter the residual when revenue replaces quantity, so price–instrument covariance becomes an omitted-variable bias.
- **Economic meaning.** The sign of the bias is ambiguous. Downward-sloping demand pushes one way; aggregate demand shocks and decreasing returns can push the other way.
- **Bottom line.** The gap between revenue- and quantity-based elasticities is governed by $\text{Cov}(p_{it}, v_{it-1})$.

3.5 Demand system and the role of price elasticities

The paper uses a log-linear approximation to a general inverse demand system:

$$p_{it} = - \sum_j \varepsilon_{ijt} y_{jt}. \quad (5)$$

- **Interpretation.** ε_{ijt} is the effect of firm j 's quantity on firm i 's price.
- **Main theoretical message.** If firms share a common production elasticity, revenue-based markups equal true markups up to a common multiplicative constant, so *dispersion* is preserved even if the *mean* is not.

In the special case where firms face heterogeneous own demand elasticities, the paper derives

$$\hat{\alpha} = \mathbb{E} \left[\alpha(1 - \varepsilon_{it}) \frac{v_{it}v_{it-1}}{\mathbb{E}[v_{it}v_{it-1}]} \right] \neq \frac{\partial r_{it}}{\partial v_{it}}, \quad (6)$$

and the resulting revenue-based markup is

$$\hat{\mu}_{it}^R = \mathbb{E} \left[\mu_{it}^{-1} \frac{v_{it}v_{it-1}}{\mathbb{E}[v_{it}v_{it-1}]} \right] \mu_{it}. \quad (7)$$

- **Interpretation.** Revenue-based markups again equal true markups times a constant.
- **Crucial implication.** The average level of the markup is not informative, but cross-firm variation is still recoverable.
- **Knife-edge failure.** If all firms have the same demand elasticity, then revenue-based markups collapse to 1, matching the Bond et al. critique.

3.6 Beyond Cobb–Douglas: translog production

The one-input translog case is

$$y_{it} = \alpha v_{it} + \beta v_{it}^2 + \omega_{it}, \quad \alpha_{it} \equiv \frac{\partial y_{it}}{\partial v_{it}} = \alpha + 2\beta v_{it}. \quad (8)$$

Revenue-based markups then satisfy

$$\hat{\mu}_{it}^R = (1 + f(v_{it}))\mu_{it}, \quad f(v_{it}) = \frac{(\hat{\alpha} - \alpha) + 2(\hat{\beta} - \beta)v_{it}}{\alpha + 2\beta v_{it}}. \quad (9)$$

- **Interpretation.** With heterogeneous output elasticities, the distortion is no longer just a common constant; it varies with input usage.
- **When things still work.** If prices are orthogonal to the instruments, the bias terms disappear and markups are well estimated.
- **Macro interpretation.** This is why the authors emphasize price–instrument orthogonality rather than insisting on literal quantity data everywhere.

3.7 Measurement error and the first-stage purging regression

With measurement error, observed output is

$$\tilde{y}_{it} = \alpha v_{it} + \omega_{it} + \eta_{it}. \quad (10)$$

Under imperfect competition, the paper shows that observed output can be written as

$$\tilde{y}_{it} = v_{it} - p_{it} + \log \mu_{it} + w_t + \eta_{it}. \quad (11)$$

- **Interpretation.** A first-stage regression that controls for input usage, prices, and objects that proxy for markups can purge measurement error before second-stage IV-GMM.
- **Paper’s implementation.** Because markups themselves are unknown, the authors use **market share** as a control, consistent with Atkeson–Burstein or Kimball-style environments where markups are tied to market share.
- **Alternative.** Blundell–Bond style one-step estimation remains consistent but can have much larger variance when measurement error is important.

3.8 Input wedges

If the flexible input faces an additional wedge λ_{it} , the measured object is not the pure markup but

$$\mu_{it}\lambda_{it} = \alpha_{it} \frac{P_{it}Y_{it}}{W_t V_{it}}. \quad (12)$$

- **Interpretation.** The production approach identifies the composite wedge on the chosen flexible input.
- **Practical lesson.** Researchers should pick the variable input least likely to be subject to additional distortions if their object of interest is the markup itself.

3.9 Simulation model

In the quantitative model, the firm's markup is an increasing function of market share:

$$\mu_{iht} = \frac{\varepsilon}{\varepsilon - 1} \left(1 - \frac{\frac{\varepsilon}{\sigma} - 1}{\varepsilon - 1} s_{iht} \right)^{-1}. \quad (13)$$

- **Interpretation.** The less substitutable are goods across broad markets relative to within-market substitution, the more market share translates into market power.

Firms produce with a translog technology:

$$y_{iht} = \omega_{iht} + \gamma \alpha v_{iht} + \gamma(1 - \alpha)k_{iht} + \gamma \frac{\alpha(1 - \alpha)\phi - 1}{2} \frac{\phi - 1}{\phi} (v_{iht}^2 + k_{iht}^2 - 2k_{iht}v_{iht}). \quad (14)$$

The estimated production function in the simulations is

$$y_{iht} = \beta_v v_{iht} + \beta_k k_{iht} + \beta_{vv} v_{iht}^2 + \beta_{kk} k_{iht}^2 + \beta_{vk} k_{iht} v_{iht} + \omega_{iht}, \quad (15)$$

with output elasticity

$$\frac{\partial y_{iht}}{\partial v_{iht}} = \beta_v + 2\beta_{vv} v_{iht} + \beta_{vk} k_{iht}. \quad (16)$$

- **Why this matters.** The simulation lets the authors compare estimated markups with true markups under realistic departures from the toy model.

3.10 Regression used to validate downstream markup relationships

In both simulations and data, the authors study whether markups behave similarly across specifications using

$$x_{it} = \chi \log \hat{\mu}_{it} + \phi_i + \psi_t + \epsilon_{it}. \quad (17)$$

- **Dependent variables.** Profit rate, material share, labor share, and market share.
- **Interpretation.** If revenue-based markups preserve the same regression relationships as quantity-based markups, then a great deal of applied work remains credible.

3.11 Empirical production function by industry

In the French data, the paper estimates a translog production function separately within each two-digit industry:

$$y_{it} = \omega_{it} + \beta_m^I m_{it} + \beta_l^I l_{it} + \beta_k^I k_{it} + \beta_o^I o_{it} + \sum_{\{h,j\} \in \{m,l,k,o\}} \beta_{hj}^I h_{it} j_{it}, \quad i \in I, \quad (18)$$

with productivity process

$$\omega_{it} = \rho^I \omega_{it-1} + \xi_{it}. \quad (19)$$

The materials elasticity is therefore

$$\alpha_{it}^m = \frac{\partial y_{it}}{\partial m_{it}} = \beta_m^I + 2\beta_{mm}^I m_{it} + \beta_{mo}^I o_{it} + \beta_{ml}^I l_{it} + \beta_{mk}^I k_{it}. \quad (20)$$

- **Interpretation.** Output elasticities differ across firms even within an industry because they depend on the firm's entire input bundle.
- **Empirical role.** These firm-specific elasticities are what feed directly into the markup formula.

3.12 Aggregate markup formula

The preferred aggregate markup is the sales-weighted harmonic mean of firm-level quantity-based markups:

$$\mu_t^{agg} = \left(\sum_{i \in I_t} s_{it} \mu_{it}^{-1} \right)^{-1}. \quad (21)$$

- **Interpretation.** This is the welfare-relevant aggregate markup in a broad class of models.
- **Comparison exercise.** The paper contrasts this benchmark with revenue-based, simple sales-weighted, and cost-weighted aggregates.

4 Identification and instruments (what makes the estimates credible)

4.1 Quantity-based identification

- **Main instrument.** Lagged flexible input usage, such as v_{it-1} , instruments current flexible input demand.
- **Exclusion restriction.** Under the baseline analytical setup, lagged input choices are outside the firm's time- t information set and hence orthogonal to current productivity shocks.
- **Relevance.** Input usage must be persistent; the paper notes that imperfect competition itself can help generate such persistence because today's competitors shape tomorrow's input demand.

- **Main concern.** With revenue data, prices contaminate the residual, so even a valid quantity instrument no longer identifies the output elasticity without bias.

4.2 What determines the revenue-data bias

- **Central object.** $\mathbb{E}[p_{it}v_{it-1}]$, or more generally the covariance between prices and production-function instruments.
- **If price–instrument covariance is low.** Revenue-based markups remain highly correlated with true markups.
- **If all firms have the same demand elasticity.** Revenue-based markups degenerate to 1 and become uninformative.
- **Therefore.** The Bond et al. critique is right about one special case and about the average level, but too strong as a general statement about dispersion and trends.

4.3 Why the two-stage estimator is modified under imperfect competition

- Standard Akerberg–Caves–Frazer style purging assumes firms are effectively price takers in the first stage.
- Under imperfect competition, the demand for the flexible input depends on more than productivity, so a first-stage regression that only partials out inputs is not enough.
- The paper’s solution is to include **price** and a **proxy for markup determinants**, namely market share, in the first stage.
- This is not a universal solution, but it is internally consistent with the class of macro models the paper is designed to discipline.

4.4 Why simulations and French data matter for credibility

- **Simulation check.** Because true markups are known, the simulations directly measure how much information survives in revenue-based estimates.
- **Empirical check.** The French data contain both quantities and prices, so the paper can compare the exact objects that macroeconomists usually observe (revenue-based markups) to the ideal benchmark (quantity-based markups).
- **Main credibility result.** In both environments, price–instrument correlations are low enough that revenue-based estimates preserve a large amount of markup variation.

4.5 Main limitations

- Revenue data are not appropriate for measuring average markup levels.
- The method can pick up input wedges in addition to pure markups.
- Empirical success relies on a sufficiently common production technology within sectors and on reasonable functional-form choices.

- When quantity measures are noisy, one-step estimators become fragile and two-step purging becomes more valuable.

5 Tables and figures

5.1 Table I: Estimated production function parameters in the simulation

Read:

- The baseline two-stage quantity estimator recovers the production function almost exactly: the estimated coefficients sit at or within a standard deviation of the truth.
- The revenue-based ACF estimator does *not* recover the structural coefficients. For example, β_v falls from 0.32 to 0.29, β_k falls from 0.48 to 0.31, and β_{kk} even flips sign.
- The key point is that coefficient bias does *not* imply useless markups. Even with biased coefficients, the elasticity relevant for markup variation can remain close enough to the truth to preserve strong correlations.

TABLE I
ESTIMATED PRODUCTION FUNCTION PARAMETERS.

	$\beta_v = \alpha\gamma$	$\beta_k = (1 - \alpha)\gamma$	$\beta_{vv} = \gamma^{\frac{\alpha(1-\alpha)}{2}} \frac{\phi-1}{\phi}$	$\beta_{kk} = \beta_{vv}$	$\beta_{vk} = -2\beta_{vv}$
True value	0.32	0.48	0.009	0.009	-0.017
Quantity (Baseline)	0.32 (0.002)	0.48 (0.001)	0.009 (0.0005)	0.009 (0.0006)	-0.018 (0.001)
Revenue (ACF)	0.29 (0.003)	0.31 (0.003)	0.005 (0.002)	-0.001 (0.004)	-0.016 (0.005)

Note: "True value" coefficients are directly calculated from the calibrated parameters (α, γ, ϕ) . "Baseline": A two-stage IV-GMM on observed quantity. "ACF": A two-stage IV-GMM on revenue. Production function coefficients are averaged across 200 Monte Carlo simulations. Standard deviations across the Monte Carlo simulations are given in parentheses.

TABLE II
OVERVIEW—MARKUP ESTIMATES.

	Correlation With True Markups	Log Markup Moments			
		Mean	St. Dev.	Median	IQR
True markups	1.00	0.23	0.07	0.21	0.09
Quantity (baseline)	1.00 (0.01)	0.22 (0.016)	0.07 (0.001)	0.21 (0.016)	0.09 (0.002)
Revenue (ACF)	0.94 (0.05)	0.23 (0.020)	0.07 (0.011)	0.21 (0.021)	0.08 (0.015)

Note: The first column presents estimates' correlations with true markups. Subsequent columns show moments of the estimated (log) markup distribution. Standard deviations across 200 Monte Carlo simulations are in parentheses. "Baseline": A two-stage IV-GMM on observed quantity. "ACF": A two-stage IV-GMM on revenue.

5.2 Table II: Overview of simulated markup estimates

Read:

- Quantity-based markups are essentially perfect: correlation with true markups is 1.00, and the moments of the log-markup distribution are almost exact.
- Revenue-based markups still perform strikingly well: the correlation with true markups is 0.94.
- The mean of log markups is almost unchanged between true and revenue-based estimates, but this should not be over-interpreted: the paper's theory says level accuracy is not generically guaranteed.

5.3 Figure 1: Simulated binned scatter of quantity- and revenue-based markups

Read:

- In levels, the binned scatter is strongly increasing with slope about 0.85.
- In first differences, the fit is even tighter, with slope about 0.99, visually close to a 45-degree line.
- This is the cleanest visual summary of the paper's core message: revenue-based markups are especially useful for studying *changes* in markups over time.

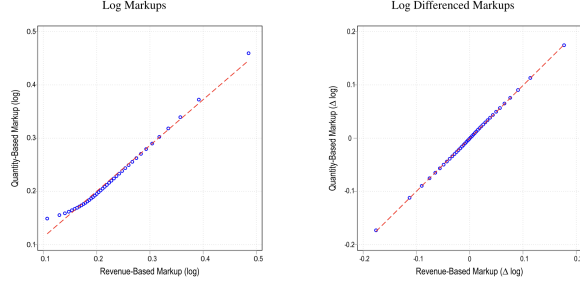


FIGURE 1.—Binned scatter plot for simulated quantity and revenue markups. *Notes:* The figures plot the relationship between quantity-based markups (“baseline”) and revenue-based markups (“ACF”) in simulated data. Log markups are used in panel (a), log differenced markups in panel (b). Linear regression coefficients are 0.85 and 0.99, respectively. The scatters are averages across Monte Carlo simulation.

TABLE III
SIMULATED RELATION BETWEEN MARKUP ESTIMATES AND OTHER VARIABLES.

	Profit Rate	Material Share	Market Share
True Markups	0.019 (0.0001)	−0.019 (0.0001)	0.068 (0.0001)
<i>R-Squared</i>	0.901	0.901	0.996
Quantity (Baseline)	0.020 (0.0001)	−0.020 (0.0001)	0.020 (0.0001)
<i>R-Squared</i>	0.899	0.899	0.995
Revenue (ACF)	0.021 (0.0001)	−0.021 (0.0001)	0.075 (0.0002)
<i>R-Squared</i>	0.902	0.902	0.967

Note: OLS coefficients. Explanatory variables are in the column headers. “Baseline”: IV-GMM on quantity. “ACF”: IV-GMM on revenue. Details in Section 4.1. Log markups are normalized to have unit standard deviations so that Δ measures the effect of a one-standard-deviation change in markups on s_{it} . Firm-clustered standard errors in parentheses. All specifications include time- and firm-fixed effects. OLS coefficients, standard errors, and R^2 's are averages across the Monte Carlo simulations.

5.4 Table III: Simulated relation between markups and other variables

Read:

- The regression coefficients linking markups to the profit rate and material share are almost identical across true, quantity-based, and revenue-based markups.
- For the profit rate, the coefficients are 0.019, 0.020, and 0.021 for true, quantity, and revenue markups, respectively.
- For the material share, the coefficients are −0.019, −0.020, and −0.021. The message is that the kinds of regressions macro papers actually run are preserved quite well under revenue-based measurement.

5.5 Figure 2: Effect of first-stage purging on markup estimates

Read:

- Without first-stage purging, the Blundell–Bond estimator deteriorates rapidly as measurement error rises.
- With first-stage purging, both quantity- and revenue-based correlations with true markups remain high even when measurement error becomes much larger than in the baseline calibration.
- The headline endpoint is important: at 16 times baseline measurement error, purging still yields correlations around 0.88 for quantity markups and 0.82 for revenue markups, while non-purged estimates fall to roughly 0.47 and 0.28.

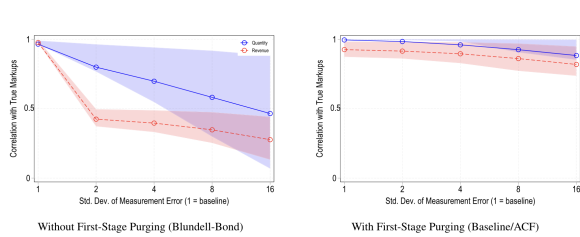


FIGURE 2.—Effect of first-stage purging regression on markup estimates. *Notes:* Figures plot the average correlation between true markups and quantity- (solid) or revenue- (dashed) based markup estimates by standard deviation of measurement error (normalized to 1 for the main calibration). Confidence intervals are the interquartile range. The left-hand panel uses the single-stage Blundell and Bond (2000) estimator; the right-hand panel uses the two-stage baseline and ACF estimator.

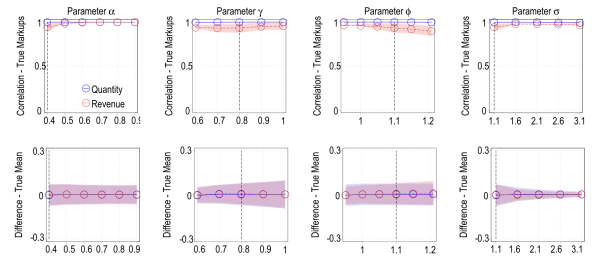


FIGURE 3.—Revenue and quantity markups for alternative parameters. *Notes:* Figures plot quantity-based (Baseline; solid) and revenue-based (ACF; dashed) markup estimates. Vertical lines give baseline values for each column's parameter. Upper figures: correlation between estimates and true markups, interquartile range in shaded areas. Bottom figures: average deviation of estimated markups' mean from true mean; shaded areas give standard deviations of that deviation across Monte Carlo simulations.

5.6 Figure 3: Robustness of the simulation results to parameter changes

Read:

- The top row shows that quantity-based markups remain essentially perfectly correlated with true markups throughout the parameter perturbations.
- Revenue-based markups also stay strong: the correlation with true markups remains at least about 0.88 even when the production and demand parameters change materially.
- The bottom row confirms the theoretical asymmetry: relative markups are robust, while deviations in the mean are the more fragile object.

5.7 Table IV: Within-sector moments of empirical log markup estimates

Read:

- In the French data, quantity-based log markups are on average larger and more dispersed than revenue-based log markups.
- The within-sector average log markup is 0.366 using quantity and 0.128 using revenue; the within-sector standard deviations are 0.232 and 0.163, respectively.
- This gap is exactly what the theory warns about: revenue data distort the level, and in real data misspecification can also compress dispersion somewhat.

TABLE IV
SUMMARY STATISTICS—WITHIN-SECTOR MOMENTS OF THE LOG MARKUP ESTIMATES.

	Mean	St. Dev.	Median*	25th Pct.*	75th Pct.*
Quantity (baseline)	0.366 (0.003)	0.232 (0.003)	−0.011 (0.004)	−0.149 (0.004)	0.137 (0.004)
Revenue (ACF)	0.128 (0.002)	0.163 (0.002)	−0.015 (0.003)	−0.113 (0.003)	0.099 (0.004)

Note: Moments are calculated at the sector level and then averaged across the sectors in the data. Parentheses present bootstrapped standard errors of the moment. “Baseline”: two-stage IV-GMM on quantity. “ACF”: two-stage IV-GMM on revenue. 147,704 obs. *: Medians and percentiles are expressed in deviation of sector averages.

TABLE V
CORRELATION BETWEEN QUANTITY- AND REVENUE-BASED LOG MARKUP ESTIMATES.

	Summary of Within-Sector Correlations					Pooled Overall
	Average	St. Dev.	Median	25th Pct.	75th Pct.	
Pearson Correlation	0.61	(0.23)	0.65	0.41	0.78	0.34
Pearson Corr. First Diff.	0.80	(0.16)	0.84	0.70	0.94	0.79
Rank Correlation	0.62	(0.23)	0.68	0.39	0.79	0.43
Rank Corr. First Diff.	0.83	(0.12)	0.84	0.71	0.95	0.84

Note: The table presents correlations between revenue- and quantity-based markup estimates. Correlations are calculated at the two-digit sector in the left-hand panel, and for all observations pooled in the right-hand panel. The summary statistics are averages of the correlations across sectors. 147,704 observations.

5.8 Table V: Correlation of quantity- and revenue-based empirical markups

Read:

- Within sectors, the Pearson correlation is 0.61 in levels and 0.80 in first differences; the corresponding rank correlations are 0.62 and 0.83.
- Median within-sector correlations are even higher than averages, which says the result is not driven by a handful of sectors.
- When all firms are pooled, level correlations fall because sectors are heterogeneous, but first-difference correlations remain high. Again, growth rates are the cleanest use case for revenue-based markups.

5.9 Figure 4: Empirical binned scatter of quantity- versus revenue-based markups

Read:

- The empirical binned scatter looks remarkably tight, especially in first differences.
- The linear-fit coefficients are about 0.89 in levels and 0.91 in first differences, which are surprisingly close to the simulation benchmarks.
- This figure is the empirical counterpart to Figure 1: even with all the messy realities of administrative data, revenue-based estimates map strongly into quantity-based estimates.

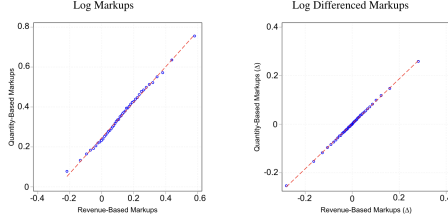


FIGURE 4.—Binned scatter plot: quantity- versus revenue-based markups. *Notes:* The figures plot the binned scatter plot between quantity-based markups (baseline) and revenue-based markups (ACF). Regression coefficients for the linear fit are 0.89 and 0.91, respectively.

TABLE VI
EMPIRICAL RELATION BETWEEN MARKUP ESTIMATES AND OTHER VARIABLES.

	Profit Rate	Labor Share	Material Share	Market Share
Quantity (baseline)	0.150 (0.002)	−0.050 (0.001)	−0.096 (0.001)	0.070 (0.006)
<i>R-Squared</i>	0.357	0.098	0.312	0.010
Revenue (ACF)	0.111 (0.001)	−0.027 (0.001)	−0.077 (0.001)	0.031 (0.004)
<i>R-Squared</i>	0.502	0.075	0.501	0.007

Note: Each entry provides the OLS estimate using the variable in the column header as the dependent variable and the markup series in the row as the regressor. Firm-clustered standard errors in parentheses. All regressions include time- and firm-fixed effects.

5.10 Table VI: Empirical relation between markup estimates and other variables

Read:

- High-markup firms are more profitable and have lower labor and material shares under both markup measures.
- Quantity-based markups load more strongly on these outcomes than revenue-based markups, but the qualitative pattern is the same. For example, the profit-rate coefficients are 0.150 and 0.111, while the material-share coefficients are −0.096 and −0.077.
- This matters for applications: many reduced-form exercises using revenue-based markups are probably directionally and often quantitatively informative.

5.11 Figure 5: Aggregate markups in France

Read:

- All series are indexed to 1 in 2010, so the figure is about *trends* rather than levels.
- Quantity-based and revenue-based aggregate markups move similarly over time: a decline around the Eurocrisis in 2011–2012 and then an upward trend.

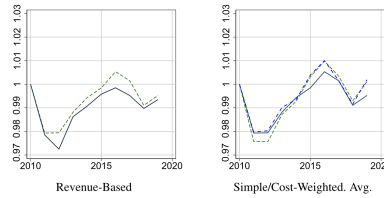


FIGURE 5.—Trend in aggregate markups in France (index = 1 in 2010). *Notes:* Solid lines plot aggregate (harmonic revenue-weighted) markups based on quantity. Dashed lines, from left to right, plot: aggregate markups based on revenue; aggregate markups based on quantity that are simple rather than harmonic weighted. Dash-dotted line in the second plot: aggregate markups based on quantity that are cost weighted. Markup levels are not shown; all are indexed to 1 in 2010.

- The levels differ substantially: average aggregate revenue-based markups are about 1.08, while quantity-based aggregate markups are about 1.45. The figure therefore perfectly summarizes the paper’s final takeaway: trends are robust, levels are not.

6 Short summary

- **Conclusion.** Financial-statement data can be used to study relative markup variation and markup trends, but not the average level of markups.
- **Intuition.** Revenue contaminates production-function estimation through unobserved prices, yet that contamination often behaves like a common or weakly varying distortion rather than a total destruction of information.
- **Empirical lesson.** The most defensible uses of revenue-based markups are within-sector dispersion, time variation, and reduced-form relationships with other firm outcomes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Revenue-based markup estimates are usually informative about true markups even when they do not recover the true production function.
- **What they identify well.** Cross-sectional dispersion, within-firm time variation, and aggregate markup trends.
- **What they do not identify well.** Average markup levels, and therefore any structural parameter that is disciplined by those levels.
- **Why the estimates work.** In both the simulations and the French data, prices are only weakly correlated with the instruments used in production-function estimation, so the revenue-data bias does not wipe out markup variation.
- **Methodological contribution.** The paper also proposes a modified first-stage purging regression for imperfect-competition settings, using price and markup proxies such as market share.

7.2 Economic intuition

- Markups are identified from a wedge between a flexible input’s elasticity and its revenue share.
- Revenue data are problematic because revenue moves with both quantity and price, but the induced elasticity bias need not mimic the revenue elasticity exactly.
- If that bias is roughly common across firms or only weakly related to the true markup, then differences in estimated markups still mirror differences in true markups.
- This is why the paper’s message is fundamentally about *which moments* of the markup distribution survive measurement problems.

7.3 Takeaway

This paper is best read as a guide for how to use accounting-data markups responsibly. It does not say revenue-based markups are perfect, and it does not reject the concerns in Bond et al. about levels. Instead, it draws the more useful distinction between objects that survive the missing-price problem and objects that do not. For macroeconomists, the practical rule is clear: use quantity data when the level matters, but do not throw away financial-statement markups when the question is about relative markups, reallocation, or trends over time.